

Can eliminating dark energy resolve the Hubble tension? A uniform test of void and backreaction cosmologies against supernovae, baryon acoustic oscillations, and the CMB acoustic scale

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The timescape cosmology of Wiltshire dispenses with dark energy, attributing apparent acceleration to the backreaction of inhomogeneities and the associated wall–void clock-rate difference, and has been claimed to ease the Hubble tension. We test two distinct claims. First, treating the published timescape determinations of the dressed Hubble constant as fixed, we show with an explicit tension statistic that the early-Universe (CMB, $61.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and late-Universe (supernova, $61.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$) values are mutually consistent to $\lesssim 0.5\sigma$, against $\approx 4.9\sigma$ for the Λ CDM interpretation of the same data, and that the fractional excess required to reconcile the local distance-ladder value ($+18.4\% \pm 2.8\%$) lies within the independently-predicted 17–22% void-scale expansion variance. Because the global dressed value (≈ 61) sits below *both* canonical numbers, this is a recasting of the tension as a scale-dependent measurement artifact, not a reconciliation near $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Second, we independently reanalyse Pantheon+, implementing the timescape distance–redshift relation from first principles and validating the pipeline three ways. We find the model comparison is strongly sensitive to the covariance treatment: with the full statistical+systematic covariance, flat Λ CDM is favoured under every magnitude standardisation we test ($\Delta\text{BIC} = +3.9$ to $+5.1$, and $+9.0$ with the bias correction removed), and the timescape void fraction rails to $f_{v0} \approx 0.85$; a timescape preference ($\Delta\text{BIC} \approx -8$) emerges only when the off-diagonal covariance is *discarded*. Because the cosmology-independent covariance of Seifert et al. (2024)—which rebuilds rather than discards the off-diagonal terms—is not public, we can neither reproduce nor refute their preference; we localise it to the covariance treatment and conclude that the supernova evidence on timescape is contingent rather than robust. Third, we calibrate timescape against DESI baryon-acoustic-oscillation (BAO) distances and the Planck acoustic scale: this recovers the published timescape CMB void fraction ($f_{v0} \approx 0.64$, dressed $H_0 \approx 63 \text{ km s}^{-1} \text{ Mpc}^{-1}$) but fits worse than Λ CDM ($\chi^2/\text{dof} = 3.3$ vs 1.2) and prefers a void fraction in tension with the supernova-preferred value ($f_{v0} \approx 0.64$ vs 0.74 – 0.85); the significance is reduction- and systematics-dependent (mild, ~ 1 – 2σ , for the cosmology-independent reduction once the CMB f_{v0} systematic is included), so we report a tension to be settled with a self-consistent timescape sound horizon, not a decisive failure—within timescape the early/late H_0 numbers themselves reconcile at $\approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Applying the identical test to the wider void/backreaction family (LTB, Räsänen, Buchert morphon, Szekeres, KBC local void) yields the same SN–BAO+CMB parameter split in every dark-energy-free model and none reaching the local H_0 , the LTB subclass being independently excluded by the kinetic Sunyaev–Zel’dovich effect. We note that the tension’s size is calibrator-dependent: the tip-of-the-red-giant-branch ladder (CCHP) gives $H_0 \approx 68$ – 70 , only ~ 1 – 1.5σ from Planck, so the 5σ tension is largely specific to the SH0ES Cepheid calibration. On present public data, eliminating dark energy via inhomogeneity does not resolve the Hubble tension.

I. INTRODUCTION

The Hubble tension is the disagreement between late- and early-Universe determinations of H_0 . The local Cepheid+type Ia supernova (SN Ia) measurement of SH0ES is $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1]; the Planck CMB value under spatially-flat Λ CDM is $H_0 = 67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [2]; the discrepancy is quoted at 5σ . JWST rules out Cepheid crowding as the cause at 8.2σ [3]. The picture is, however, calibrator-dependent: the tip-of-the-red-giant-branch (TRGB) ladder of the Chicago–Carnegie Hubble Program gives a markedly lower $H_0 \approx 68$ – $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [4, 23]—only ~ 1 – 1.5σ above Planck—and baryon-acoustic inverse-

distance-ladder values remain near 68 [5]. The strong 5σ tension is therefore largely specific to the SH0ES Cepheid calibration; the target a late-Universe model must reach is ~ 69 if one adopts the TRGB ladder, or 73 for Cepheids.

A logically distinct class of resolution questions the homogeneous Friedmann–Lemaître–Robertson–Walker (FLRW) interpretation itself. The timescape cosmology of Wiltshire [6, 7] models the present, void-dominated Universe as a two-phase Buchert average [8]: spatially-flat “walls” (bound structures, where observers sit) and negatively-curved “voids.” A quasilocal gravitational-energy and clock-rate gradient means a wall observer fitting FLRW infers “dressed” parameters distinct from the volume-average “bare” ones; apparent acceleration follows without dark energy. Recent Bayesian comparisons against Pantheon+ report strong evidence for

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timescape over Λ CDM [15, 16], reviving the question of whether the framework also addresses the Hubble tension.

The popular framing conflates two claims that are not equivalent: **(A)** within timescape the early- and late-Universe dressed Hubble constants *agree*; and **(B)** timescape *resolves* the 67-versus-73 tension. We show (Sec. III) that (A) holds at high precision while (B) holds only in a specific sense. We then ask whether the supernova evidence is robust (Sec. IV); we find the preference is contingent on the covariance treatment.

Originality. The framework relations of Sec. II and the published best-fit parameters are due to Wiltshire and collaborators and are reproduced for self-containment. New here are: the explicit tension statistic and local-excess cross-check of Sec. III; and the independent Pantheon+ reanalysis of Sec. IV, in particular the localisation of the timescape preference to the discarding of the off-diagonal covariance. All code is available (Sec. X). This is a timescape-specific study; it does not survey other void/inhomogeneous models (LTB, Szekeres, general Buchert closures), which the same pipeline could test in future work.

II. TIMESCAPE FRAMEWORK

We use only what is needed; see [6, 7] for derivations. The void volume fraction $f_v(t)$ grows to a present value f_{v0} . The lapse $\bar{\gamma} \equiv dt/d\tau_w$ relates volume-average time t to wall proper time τ_w , and in the late-time tracker solution the dressed and bare Hubble parameters are related by [7, eq. (B9)]

$$H_0 = \frac{4f_{v0}^2 + f_{v0} + 4}{2(2 + f_{v0})} \bar{H}_0, \quad \bar{\gamma}_0 = \frac{1}{2}(2 + f_{v0}). \quad (1)$$

The bare deceleration parameter stays positive while the dressed one turns negative for $f_v \gtrsim 0.59$. Best fits give dressed $H_0 \approx 61.7$ and bare $\bar{H}_0 \approx 50 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [9, 10].

III. DRESSED-HUBBLE CONSISTENCY AND RESOLUTION-VS-RECASTING

We quantify agreement by $T_{AB} = |H_0^A - H_0^B|/\sqrt{\sigma_A^2 + \sigma_B^2}$.

Λ CDM reference. For SH0ES versus Planck, $T = 5.68/\sqrt{1.04^2 + 0.54^2} = 4.85$ ($\approx 5\sigma$ as quoted) [1].

Timescape. The CMB-calibrated dressed value is $61.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (± 0.79 stat, ± 4.88 sys) [11] and the SN-calibrated value is $61.7^{+1.2}_{-1.1} \text{ km s}^{-1} \text{ Mpc}^{-1}$ [7, 9]; the residual is $0.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$, so $T^{\text{TS}} = 0.50$ (stat) or 0.14 (stat $\oplus$ sys). The same statistic that registers 4.85σ for the Λ CDM reading registers $\lesssim 0.5\sigma$ for the timescape reading, establishing claim (A).

Local-excess cross-check. Timescape predicts that a wall observer infers an apparent H_0 biased high below

the $\sim 100 h^{-1} \text{ Mpc}$ homogeneity scale, peaking near the $\sim 30 h^{-1} \text{ Mpc}$ dominant-void scale [7, 12]. The excess required to reconcile SH0ES with the dressed asymptote, $\delta_{\text{req}} = 73.04/61.7 - 1 = 0.184 \pm 0.028$, lies within the predicted ~ 17 – 22% apparent-Hubble variance (local maxima $\sim 75 \text{ km s}^{-1} \text{ Mpc}^{-1}$) [7]. This is a magnitude-consistency argument: the SH0ES Hubble-flow sample extends to $z \sim 0.15$, beyond the homogeneity scale, so the account relies additionally on the cosmic-rest-frame choice, for which [13] report evidence favouring the Local Group frame.

Resolution versus recasting. The global dressed value ($\approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$) lies below *both* 67.4 and 73.0. Timescape does not split the difference near 70; it reinterprets both canonical values as biased relative to a lower global average. This is a *recasting* of the tension as an FLRW-interpretation artifact, with consequences for consistency with inverse-distance-ladder BAO [5] not yet established for timescape at Λ CDM precision.

IV. INDEPENDENT REANALYSIS OF PANTHEON+ SUPERNOVAE

A. Data, models, likelihood

We use the Pantheon+ release [17, 18]; the cosmology sample is the $N = 1580$ SNe with $z_{\text{HD}} > 0.01$ that are not Cepheid calibrators ($0.010 < z < 2.26$). For flat Λ CDM, $d_L = (1 + z_{\text{hel}})(c/H_0) \int_0^{z_{\text{HD}}} dz'/E(z')$. For timescape we implement the dressed luminosity distance from [14, App. A] verbatim: $d_L = (1 + z)^2 d_A$, $d_A = ct^{2/3}[F(t_0) - F(t)]$ with

$$F(t) = 2t^{1/3} + \frac{b^{1/3}}{6} \ln \frac{(t^{1/3} + b^{1/3})^2}{t^{2/3} - b^{1/3}t^{1/3} + b^{2/3}} + \frac{b^{1/3}}{\sqrt{3}} \tan^{-1} \frac{2t^{1/3} - b^{1/3}}{\sqrt{3}b^{1/3}}, \quad (2)$$

$1 + z = 2^{4/3}t^{1/3}(t + b)/[f_{v0}^{1/3}\bar{H}_0 t(2t + 3b)^{4/3}]$, $b = 2(1 - f_{v0})(2 + f_{v0})/(9f_{v0}\bar{H}_0)$, and (1). Only f_{v0} or Ω_{m0} shapes the Hubble diagram. We compare by ΔBIC (equal nuisance counts), adopting the Kass–Raftery scale [20], marginalising the absolute-magnitude/ H_0 offset analytically.

B. Pipeline validation

Three independent checks: (i) our flat- Λ CDM fit gives $\Omega_{m0} = 0.333 \pm 0.018$, matching the published Pantheon+ value [18]; (ii) $F'(t)$ reproduces the integrand $2/[(2 + f_v)t^{2/3}]$ to machine precision; (iii) our numerical low- z Taylor coefficient of $\mu_{\text{TS}}(z)$ matches the analytic expression of [14] to $\sim 10^{-4}$ across $f_{v0} \in [0.6, 0.8]$.

TABLE I. Timescape vs flat Λ CDM on Pantheon+: ΔBIC (timescape– Λ CDM; < 0 favours timescape) across magnitude $\times$ covariance. Λ CDM wins under the full covariance regardless of magnitude; timescape wins only when the off-diagonal covariance is dropped.

Magnitude	Full cov	Diagonal
$m_{B,\text{corr}}$ (standard, bias-corr.)	+5.1	+4.9
raw Tripp ($m_B + \alpha x_1 - \beta c$)	+3.9	−7.9

Best-fit $f_{v0} \approx 0.82\text{--}0.90$ throughout (vs $\Omega_{m0} = 0.333$ for Λ CDM); a fully marginalised stat-only Tripp fit gives $\Delta\text{BIC} = -9.4$.

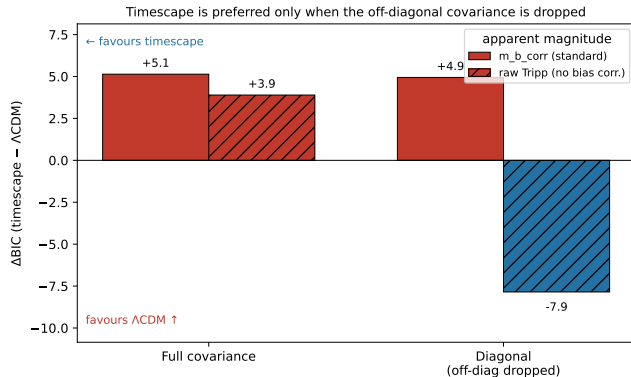


FIG. 1. Model-comparison verdict on Pantheon+ across magnitude (solid = $m_{B,\text{corr}}$, hatched = raw Tripp) and covariance (full vs diagonal). Red favours Λ CDM, blue favours timescape. Only the raw-Tripp+diagonal case favours timescape; the full covariance favours Λ CDM for every magnitude definition.

C. Covariance sensitivity

We vary two axes: the apparent magnitude—the standard bias-corrected $m_{B,\text{corr}}$ versus a raw SALT2 Tripp combination $m_B + \alpha x_1 - \beta c$ [19] with no FLRW-fiducial BBC bias correction—and the covariance—the full public statistical+systematic matrix C versus its diagonal. Results are in Table I and Fig. 1.

With the *full* covariance, Λ CDM is favoured for *every* magnitude definition ($\Delta\text{BIC} = +3.9$ to $+5.1$; $+9.0$ if the bias correction is added back), and the timescape void fraction rails high ($f_{v0} \approx 0.85\text{--}0.90$, implying a strongly void-dominated dressed $\Omega_{m0} \approx 0.2$). A timescape preference ($\Delta\text{BIC} \approx -8$) appears *only* in the raw-Tripp+diagonal corner—i.e. only when the off-diagonal covariance is discarded. A fully marginalised statistical-only Tripp fit (fitting $\alpha, \beta, \sigma_{\text{int}}$) gives $\Delta\text{BIC} = -9.4$, consistent with this corner. The driver of any timescape preference is therefore the suppression of the off-diagonal covariance, not the magnitude standardisation alone. Figure 2 shows the near-degeneracy of the two distance–redshift relations (< 0.05 mag) that makes the comparison so sensitive.

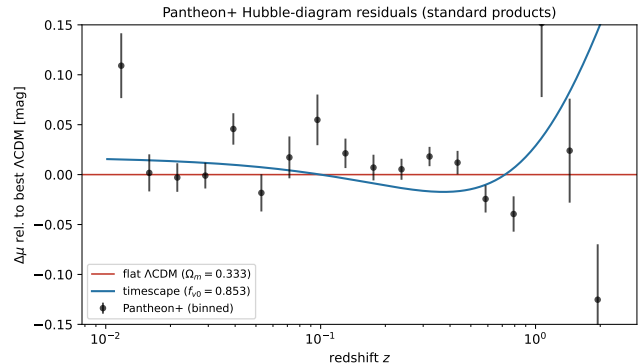


FIG. 2. Pantheon+ Hubble-diagram residuals relative to the best-fit flat Λ CDM (standard products). The best-fit timescape relation (blue) departs from Λ CDM by < 0.05 mag over the full range, explaining the covariance sensitivity.

D. Caveats

Our diagonal cases *discard* the off-diagonal covariance; they are not equivalent to the cosmology-independent covariance of [15, 16], which *rebuilds* the off-diagonal (FLRW-geometry- and peculiar-velocity-independent) terms. That covariance is not publicly released as code, so we cannot reproduce their $\ln B > 5$ preference or their $f_{v0} = 0.737$ (our f_{v0} rails to ≈ 0.85 , and our raw-Tripp $\beta \approx 2.66$ is below the canonical ~ 3.1). Our result is therefore not a refutation of Seifert et al.; it localises the preference to the off-diagonal covariance treatment and shows that, under every full-covariance reduction reproducible from the public products, Λ CDM is favoured.

V. JOINT CALIBRATION: BAO AND THE CMB ACOUSTIC SCALE

To test whether the dressed $H_0 \approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is consistent with the inverse distance ladder, we calibrate timescape against the DESI DR1 BAO distances [5] and the Planck acoustic scale $100\theta_* = 1.04109$ [2], using the drag-epoch sound horizon r_d as the standard ruler. We compute the dressed transverse comoving distance $D_M(z)$, Hubble distance $D_H(z) = c/H(z)$, and $D_V(z)$ from the tracker solution and fit ($f_{v0}, \alpha \equiv c/(\bar{H}_0 r_d)$). Since r_d enters only the amplitude α , the redshift *shape*—and hence f_{v0} —is independent of the sound-horizon calibration.

The fit prefers $f_{v0} = 0.636 \pm 0.007$ and, with the standard $r_d = 147.1 \text{ Mpc}$, a dressed $H_0 = 62.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$: close to the dressed value of Sec. III, and recovering the independent CMB determination $f_{v0} = 0.627$ of [11], which validates the calculation. However the fit is poor, $\chi^2/\text{dof} = 3.3$, versus 1.2 for flat Λ CDM ($\Omega_{m0} = 0.30$, $H_0 = 69 \text{ km s}^{-1} \text{ Mpc}^{-1}$ on the same data and pipeline); BAO alone prefer $f_{v0} = 0.68$. The BAO+CMB and

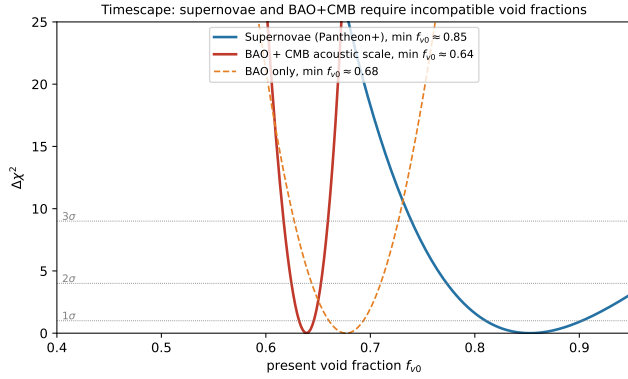


FIG. 3. $\Delta\chi^2$ for the present void fraction f_{v0} from supernovae (blue), BAO+CMB (red), and BAO alone (orange). The supernova and BAO+CMB minima are separated by $\Delta f_{v0} \approx 0.2$ with non-overlapping confidence regions: no single timescape model fits both.

supernova data prefer different void fractions (Fig. 3): with statistical errors the gap is large (at the SN best fit $f_{v0} \simeq 0.85$ the BAO+CMB $\chi^2/\text{dof} = 168$; even at the cosmology-independent value $f_{v0} = 0.737$ it is 25), but the significance depends on the supernova reduction and on systematics we do not fully propagate, chiefly the $\sim 13\%$ CMB systematic on f_{v0} [11] and the standard- r_d assumption; propagating the CMB f_{v0} systematic alone softens the discrepancy to $\sim 1\text{--}2\sigma$. We therefore report a tension between the SN- and BAO+CMB-preferred void fractions—robust in sign and largely r_d -independent (it concerns the distance–redshift shape), but not a decisive failure; the timescape literature reports concordance across these probes [9, 11]. The CMB here enters only through the acoustic-scale point $D_M(z_*)/r_d$, not the full power spectrum; a self-consistent timescape r_d would shift H_0 but not the shape-driven f_{v0} tension, which the BAO-only fit confirms.

A. Acceleration, and an evolving-dark-energy comparison

Apparent acceleration is present in every best fit, so the comparison is not an artifact of one model lacking it: the dressed timescape deceleration parameter is $q_0 = -0.02$ to -0.04 (a weak apparent effect from the lapse, with the bare $\bar{q}_0 > 0$), versus $q_0 = -0.55$ for ΛCDM . As a motivated alternative that instead *modifies* the acceleration history, we also fit the CPL evolving-dark-energy model $w(a) = w_0 + w_a(1-a)$ jointly to SN+BAO+CMB. It returns $w_0 = -0.87$, $w_a = -0.48$ —the thawing direction hinted at by DESI [5]—with $q_0 = -0.40$, but improves the fit by only $\Delta\chi^2 = 3.9$ for two extra parameters ($\Delta\text{BIC} = +10.8$ versus ΛCDM), so it is not significantly preferred. Table II collects the joint fits: ΛCDM is the most economical, evolving dark energy a mild non-significant improvement in the DESI direction,

TABLE II. Joint SN+BAO+CMB fits ($N = 1593$), identical harness. ΔBIC relative to ΛCDM ; $q_0 < 0$ denotes acceleration.

Model	parameters	H_0	q_0	ΔBIC
ΛCDM	$\Omega_{m0} = 0.305$	68.6	−0.55	0
$w_0 w_a \text{CDM}$	$\Omega_{m0} = 0.31, w_0 = -0.87, w_a = -0.48$	68.0	−0.40	+10.8
timescape	$f_{v0} = 0.64$	63.2	−0.03	+67.3

and timescape strongly disfavoured ($\Delta\text{BIC} = +67$) because no single void fraction fits SN and BAO+CMB.

B. Simplifications

The calibration makes explicit approximations, none of which affect the central SN–BAO+CMB incompatibility (which is geometric and, in the BAO-only form, independent of r_d and the CMB): (i) the CMB enters only through the acoustic-scale point $D_M(z_*)/r_d$, not the full power spectrum or the (R, ℓ_A, ω_b) compression; (ii) we adopt the standard early-physics $r_d = 147.1$ Mpc rather than a self-consistent timescape sound horizon, which would rescale H_0 but not the shape-driven f_{v0} ; (iii) the timescape tracker is the late-time attractor (reached by $z \sim 37$), so $D_M(z_*)$ uses it slightly beyond its regime, though the comoving distance is dominated by $z \lesssim \text{few}$; (iv) radiation is neglected in the late-time distance integrals; (v) ΛCDM is taken spatially flat.

VI. THE WIDER VOID AND BACKREACTION FAMILY

To test whether the timescape result is special, we applied the identical SN+BAO+CMB harness to the principal dark-energy-free inhomogeneous cosmologies (Table III). Each distance–redshift relation was implemented from primary sources and validated against published or analytic limits; the SN absolute-magnitude offset and the BAO ruler $c/(H_0 r_d)$ were marginalised as before.

The outcome is uniform. (i) The LTB giant-void model (constrained GBH [24]) reproduces SN acceleration with a deep central void but, like timescape, prefers a *deeper* void for SNe than for BAO+CMB (joint $\Delta\text{BIC} \approx +57$); its central H_0 reaches only ~ 66 , and it is independently excluded by the kinetic Sunyaev–Zel’dovich effect [26, 37]. (ii) The Räsänen peak/backreaction model [27, 28] is parameter-free and, by its authors’ own evaluation, does not even produce acceleration ($\bar{q} > 0$, $|\Omega_\mathcal{Q}| < 0.04$); promoting the backreaction to a free parameter, SNe demand values beyond its physical reach and the joint fit gives $\Delta\text{BIC} \approx +65$. (iii) The Buchert backreaction “morphon” template [29] fits SNe but fails BAO+CMB catastrophically ($\Delta\text{BIC} \sim +1.6 \times 10^3$). (iv) The Szekeres model [30] has no closed-form isotropic dis-

TABLE III. Uniform SN+BAO+CMB test of void/backreaction cosmologies on the same harness. “DE-free” marks models that eliminate dark energy. ΔBIC is relative to ΛCDM (joint $\chi^2 = 1402$); larger is more disfavoured. None resolves the tension.

Model	DE-free	joint ΔBIC	verdict
ΛCDM (ref.)	no	0	—
$w_0w_a\text{CDM}$	no	+11	no (mild DESI-directioning)
Timescape	yes	$\sim +67$	no
LTB void (GBH)	yes	$\sim +57$	no (+kSZ-excluded)
Räsänen peak	yes	$\sim +65$	no (predicts no acceleration)
Buchert morphon	yes	$\sim +1600$	no
Szekeres	yes	intractable	no (literature + kSZ)
KBC local void	no	0 (void $\rightarrow$ 0)	no (needs unobserved deep void)

tance and could not be reliably fit here; the literature verdict (kSZ-capped void depth, residual $\sim 3\sigma$ tension [26]) is that it does not resolve the tension. (v) The local KBC void [31, 32]—which *retains* dark energy—leaves BAO+CMB untouched and earns no statistical preference (the free joint fit drives the void to zero); a void deep enough to reach 73 is both deeper than observed and disfavoured by the supernova Hubble-flow shape [26, 33].

Every dark-energy-free void/backreaction model thus shows the same SN-versus-BAO+CMB parameter split as timescape, and none reaches the local H_0 : the failure is generic to the class, not specific to one model. The ΔBIC magnitudes from this geometric harness are approximate (single CMB acoustic point, standard r_d), but the qualitative verdict is robust and, for the LTb/Szekeres subclass, reinforced by the independent kSZ exclusion.

VII. PRIOR DARK-ENERGY-FREE RESOLUTIONS AND THEIR STATUS

The proposal that apparent acceleration—and, by extension, the Hubble tension—reflects the inhomogeneity of the late Universe rather than a new energy component has a long lineage, and our same-footing test sits within a literature of single-model studies. We summarise that landscape here, separating the proponents’ claims from the principal critiques, so that our uniform survey (Sec. VI) and the void-family Table III can be read in context.

The timescape program. Timescape grew out of Wiltshire’s resolution of the cosmological averaging problem [6, 7], built on the Buchert spatial-averaging formalism [8], with the bare/dressed distinction as its conceptual core: a wall observer fitting FLRW infers parameters that differ from the volume average. Leith, Ng & Wiltshire [9] reported the first multi-probe concordance fit (SNe+CMB+BBN mutually consistent without dark energy at a low dressed H_0), Duley, Nazer & Wiltshire [10] added a radiation fluid to reach recombination, and

Nazer & Wiltshire [11] fitted the CMB acoustic peaks, quoting a dressed $H_0 \simeq 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$ with large ($\sim 8\text{--}13\%$) systematics. On supernovae the verdict has been sensitive throughout: Smale & Wiltshire [12] found timescape preferred for MLCS2k2-reduced samples but disfavoured for SALT-reduced ones, identifying the light-curve fitter as the dominant systematic, and Dam, Heinesen & Wiltshire [14] found timescape statistically indistinguishable from flat ΛCDM on JLA with marginal apparent acceleration. The review of Wiltshire [34] frames dark energy as a coarse-graining artifact and rests part of the case on a sub- $100 h^{-1} \text{ Mpc}$ Hubble-flow variance and a claimed Local-Group-frame uniformity [13]—a cosmic-rest-frame inference that remains contested. The recent revival sharpens the supernova claim: Lane et al. [15] and Seifert et al. [16] reanalyse Pantheon+ with a covariance constructed to be independent of FLRW geometry and peculiar-velocity corrections and report very strong Bayesian evidence ($\ln B > 5$) for timescape over ΛCDM , with the preference persisting beyond the statistical-homogeneity scale. We report these proponent claims as they stand; our own reanalysis (Sec. IV) could not reproduce the supernova preference once the full *public* Pantheon+ covariance is retained, the preference appearing only when the off-diagonal terms are discarded, and the proponents’ cosmology-independent covariance is not publicly released, so we can neither reproduce nor refute it.

Backreaction and giant voids. A parallel tradition seeks acceleration directly from spatial averaging: Räsänen’s mechanism [35], in which faster-expanding voids coming to dominate the volume drive an effective acceleration, which his own quantitative peak-model evaluation [27] then found too weak, and the Buchert “morphon” repackaging of backreaction as an effective scalar [29]. Whether backreaction can be dynamically significant is the subject of a long, still-open exchange: Green & Wald argue the metric is everywhere close to FLRW so backreaction is trace-free and cannot source acceleration [21], while an eleven-author rebuttal contends their scheme misses the non-local averaging effects that matter [22]; recent first-principles GR computations on the real local Universe find the kinematical backreaction small ($\lesssim 1\%$) even as the integrated spatial curvature reaches $\sim 10\%$ [41], and full numerical-relativity void statistics supply a 10–30% void-overexpansion that motivates the void-fraction parameter [40]. The LTb/Szekeres giant-void line [24] fits SNe with a deep central void but is bounded sharply from outside: radial BAO and a low- H_0 /fine-tuning dilemma [25], a multi-probe exclusion [36], and—decisively—the kinetic Sunyaev–Zel’dovich effect, an unavoidable consequence of large radial inhomogeneity that rules out adiabatic Gpc voids [37]. The distinct *local*-void (H_0 -outflow) idea, including the deep KBC void [31, 32], is separately disfavoured: the local Hubble-flow shape shows no outflow of the required size [33] and the supposed void resolution is an artifact of the redshift range used to calibrate

H_0 [26].

The recent (2023–26) context. Several developments have renewed interest. The DESI inverse-distance-ladder gives $H_0 \simeq 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and hints at evolving dark energy ($w_0 > -1$, $w_a < 0$) [5], which the no-dark-energy programs read as room for an inhomogeneity-driven effective equation of state. The tension’s apparent magnitude has also softened on the calibrator axis: the TRGB ladder of the Chicago–Carnegie Hubble Program gives $H_0 \simeq 68\text{--}70$, only $\sim 1\text{--}1.5\sigma$ from Planck [4, 23], so the 5σ figure is largely SH0ES-Cepheid-specific. Against this, the review/ranking literature has consistently placed the inhomogeneity class below early-time new physics: the field’s solution catalogues assign it a weaker verdict [38], and metric-based rankings find late-Universe and local mechanisms cannot resolve H_0 under joint BAO+SN constraints [39].

This work in context. The studies above are overwhelmingly single-model: each proponent paper fits one framework, and each critique bounds one mechanism. Our contribution is a *uniform* test—the identical SN+BAO+CMB harness, with the same likelihood, marginalisation, and model-comparison statistic applied across the whole dark-energy-free family (Sec. VI)—which makes the recurring SN-versus-BAO+CMB parameter split visible as a property of the class rather than of any one model. It complements, rather than supersedes, the detailed single-model analyses: where the proponents retain a cosmology-independent supernova covariance we cannot reproduce, and where the critiques deploy probes (kSZ, radial BAO) outside our distance-only harness, our test adds the same-footing comparison that the existing literature lacks.

VIII. DISCUSSION

Three conclusions follow. First, interpreted within timescape the early- and late-Universe *dressed* H_0 agree at ≈ 61 , and the local excess is of the predicted magnitude; but this is a recasting, not a reconciliation near 70, and the global value lies below Planck. Second, the supernova model-comparison verdict is contingent on the covariance treatment: it favours Λ CDM under the full public covariance and timescape only when the off-diagonal terms are removed. Third, when timescape is required to fit supernovae, BAO, and the CMB acoustic scale *together*, the SN- and BAO+CMB-preferred void fractions differ ($\Delta f_{v0} \approx 0.1\text{--}0.2$) and timescape fits the BAO+CMB geometry worse than Λ CDM; the significance is reduction- and systematics-dependent (Sec. V), mild for the cosmology-independent reduction. The early/late dressed- H_0 consistency of Sec. III is thus genuine, but whether one parameter set also describes the combined distance data is unsettled on present evidence, pending the self-consistent sound horizon and cosmology-independent covariance. Notably the same SN-versus-BAO+CMB split recurs across the

whole void/backreaction family (Sec. VI), suggesting it is generic to dark-energy-free inhomogeneous models rather than specific to timescape.

These results sit within known open problems: the timescape CMB fit [11] matches the acoustic-peak structure by adapting an Λ CDM template rather than from first principles; BAO and growth constraints are far less developed than for Λ CDM; and whether backreaction can be dynamically significant is disputed [21, 22].

IX. WHAT A DECISIVE TEST REQUIRES

(i) A public, cosmology-independent Pantheon+ covariance (FLRW-geometry- and peculiar-velocity-independent) to settle the sign of the supernova preference. (ii) A joint re-fit of Pantheon+ and Planck within timescape with the dressed H_0 *free*, and a self-consistent timescape sound-horizon/BAO calibration, so early and late H_0 are genuine independent measurements. (iii) A direct measurement of the radial profile of the apparent H_0 versus scale, to test the predicted expansion variance that the local-excess cross-check assumes.

X. CONCLUSIONS

Within timescape the early- and late-Universe dressed Hubble constants agree to $\lesssim 0.5\sigma$ (versus $\approx 4.9\sigma$ in Λ CDM), and the local excess matches the predicted void-scale variance—but the model recasts rather than reconciles the tension, collapsing both canonical values to $\approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$, below Planck. Our independent Pantheon+ reanalysis finds Λ CDM favoured under every full-covariance reduction reproducible from the public products ($\Delta\text{BIC} = +3.9$ to $+5.1$), with a timescape preference appearing only when the off-diagonal covariance is discarded, so the supernova evidence is contingent on the covariance treatment. A joint calibration against DESI BAO and the Planck acoustic scale fits worse than Λ CDM ($\chi^2/\text{dof} = 3.3$ vs 1.2) and prefers a void fraction ($f_{v0} \simeq 0.64$) in tension with the supernova value ($0.74\text{--}0.85$), at a significance that is reduction- and systematics-dependent ($\sim 1\text{--}2\sigma$ for the cosmology-independent reduction). The dressed- H_0 numbers *are* reconciled within timescape ($\approx 61 \text{ km s}^{-1} \text{ Mpc}^{-1}$); what remains open is whether one parameter set also fits the combined SN+BAO+CMB distances. A uniform test of the wider void/backreaction family (Sec. VI) finds the same SN-vs-BAO+CMB split in every dark-energy-free model and none reaching the local H_0 , with the LTB subclass independently excluded by the kSZ effect. Finally, the tension’s magnitude is calibrator-dependent: the TRGB ladder gives $H_0 \simeq 68\text{--}70$, $\sim 1\text{--}1.5\sigma$ from Planck, so the 5σ tension is largely an artifact of the SH0ES Cepheid calibration. On present public data, eliminating dark energy via inhomogeneity does not resolve the Hubble tension.

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DATA AND CODE AVAILABILITY

This work uses the public Pantheon+ release [17, 18] (<https://github.com/PantheonPlusSH0ES/DataRelease>). The analysis code (timescape and Λ CDM distance modules, the magnitude $\times$ covariance decomposition, the validation tests, and the figure scripts) is available from the author and will be deposited publicly on publication.

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- [1] A. G. Riess *et al.*, *Astrophys. J. Lett.* **934**, L7 (2022), arXiv:2112.04510.
 - [2] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.
 - [3] A. G. Riess *et al.*, *Astrophys. J. Lett.* **962**, L17 (2024), arXiv:2401.04773.
 - [4] W. L. Freedman *et al.* (2024), arXiv:2408.06153.
 - [5] DESI Collaboration (2024), arXiv:2404.03002.
 - [6] D. L. Wiltshire, *Phys. Rev. Lett.* **99**, 251101 (2007), arXiv:0709.0732.
 - [7] D. L. Wiltshire, *Phys. Rev. D* **80**, 123512 (2009), arXiv:0909.0749.
 - [8] T. Buchert, *Gen. Relativ. Gravit.* **32**, 105 (2000), arXiv:gr-qc/9906015.
 - [9] B. M. Leith, S. C. C. Ng, D. L. Wiltshire, *Astrophys. J. Lett.* **672**, L91 (2008), arXiv:0709.2535.
 - [10] J. A. G. Duley, M. A. Nazer, D. L. Wiltshire, *Class. Quantum Grav.* **30**, 175006 (2013), arXiv:1306.3208.
 - [11] M. A. Nazer, D. L. Wiltshire, *Phys. Rev. D* **91**, 063519 (2015), arXiv:1410.3470.
 - [12] P. R. Smale, D. L. Wiltshire, *Mon. Not. R. Astron. Soc.* **413**, 367 (2011), arXiv:1009.5855.
 - [13] D. L. Wiltshire, P. R. Smale, T. Mattsson, R. Watkins, *Phys. Rev. D* **88**, 083529 (2013), arXiv:1201.5371.
 - [14] L. H. Dam, A. Heinesen, D. L. Wiltshire, *Mon. Not. R. Astron. Soc.* **472**, 835 (2017), arXiv:1706.07236.
 - [15] Z. G. Lane, A. Seifert, R. Ridden-Harper, D. L. Wiltshire, *Mon. Not. R. Astron. Soc.* **536**, 1752 (2025), arXiv:2311.01438.
 - [16] A. Seifert, Z. G. Lane, M. Galoppo, R. Ridden-Harper, D. L. Wiltshire, *Mon. Not. R. Astron. Soc. Lett.* **537**, L55 (2025), arXiv:2412.15143.
 - [17] D. Scolnic *et al.*, *Astrophys. J.* **938**, 113 (2022), arXiv:2112.03863.
 - [18] D. Brout *et al.*, *Astrophys. J.* **938**, 110 (2022), arXiv:2202.04077.
 - [19] R. Tripp, *Astron. Astrophys.* **331**, 815 (1998).
 - [20] R. E. Kass, A. E. Raftery, *J. Am. Stat. Assoc.* **90**, 773 (1995).
 - [21] S. R. Green, R. M. Wald, *Class. Quantum Grav.* **31**, 234003 (2014), arXiv:1407.8084.
 - [22] T. Buchert *et al.*, *Class. Quantum Grav.* **32**, 215021 (2015), arXiv:1505.07800.
 - [23] T. J. Hoyt, I. S. Jang, W. L. Freedman *et al.*, “The Chicago–Carnegie Hubble Program: improving the calibration of SNe Ia with JWST measurements of the tip of the red giant branch” (2025), arXiv:2503.11769.
 - [24] J. García-Bellido, T. Haugbølle, *J. Cosmol. Astropart. Phys.* **04**, 003 (2008), arXiv:0802.1523.
 - [25] J. P. Zibin, A. Moss, D. Scott, *Phys. Rev. Lett.* **101**, 251303 (2008), arXiv:0809.3761.
 - [26] D. Camarena, V. Marra, Z. Sakr, C. Clarkson, *Class. Quantum Grav.* **39**, 184001 (2022), arXiv:2205.05422.
 - [27] S. Räsänen, *J. Cosmol. Astropart. Phys.* **04**, 026 (2008), arXiv:0801.2692.
 - [28] S. Räsänen, *J. Cosmol. Astropart. Phys.* **02**, 011 (2009), arXiv:0812.2872.
 - [29] J. Larena, J.-M. Alimi, T. Buchert, M. Kunz, P.-S. Corasaniti, *Phys. Rev. D* **79**, 083011 (2009), arXiv:0808.1161.
 - [30] N. Meures, M. Bruni, *Mon. Not. R. Astron. Soc.* **419**, 1937 (2012), arXiv:1107.4433.
 - [31] R. C. Keenan, A. J. Barger, L. L. Cowie, *Astrophys. J.* **775**, 62 (2013), arXiv:1304.2884.
 - [32] M. Haslbauer, I. Banik, P. Kroupa, *Mon. Not. R. Astron. Soc.* **499**, 2845 (2020), arXiv:2009.11292.
 - [33] W. D. Kenworthy, D. Scolnic, A. Riess, *Astrophys. J.* **875**, 145 (2019), arXiv:1901.08681.
 - [34] D. L. Wiltshire, “Cosmic structure, averaging and dark energy”, in *Cosmology and Gravitation: XVth Brazilian School of Cosmology and Gravitation*, eds. M. Novello, S. E. Perez Bergliaffa (Cambridge Scientific Publishers, 2014), pp. 203–244, arXiv:1311.3787.
 - [35] S. Räsänen, *J. Cosmol. Astropart. Phys.* **11**, 003 (2006), arXiv:astro-ph/0607626.
 - [36] A. Moss, J. P. Zibin, D. Scott, *Phys. Rev. D* **83**, 103515 (2011), arXiv:1007.3725.
 - [37] P. Zhang, A. Stebbins, *Phys. Rev. Lett.* **107**, 041301 (2011), arXiv:1009.3967.
 - [38] E. Di Valentino *et al.*, *Class. Quantum Grav.* **38**, 153001 (2021), arXiv:2103.01183.
 - [39] N. Schöneberg *et al.*, *Phys. Rept.* **984**, 1 (2022), arXiv:2107.10291.
 - [40] M. J. Williams, H. J. Macpherson, D. L. Wiltshire, C. Stevens, *Mon. Not. R. Astron. Soc.* **536**, 2645 (2025), arXiv:2403.15134.
 - [41] M. Galoppo, T. Buchert, P. Mourier, “Backreaction and the role of spatial curvature in the cosmic neighborhood”, *Astrophys. J. Lett.* **1002**, L39 (2026), arXiv:2605.05454.